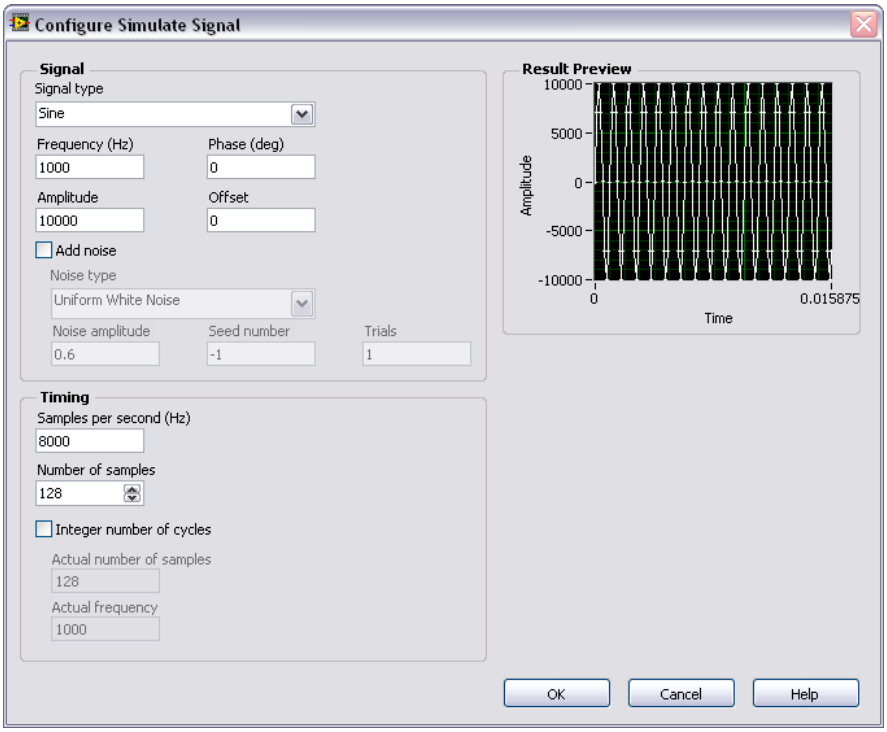
Generarea, afisarea si filtrarea semnalelor in LabVIEW

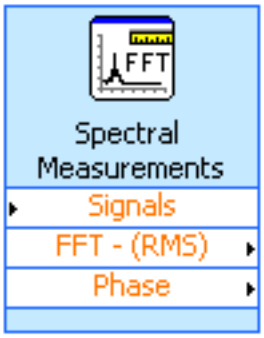
Se afiseaza diagrama bloc si din Functions palette se introduce in diagrama bloc VI-ul Simulate Signal.b:



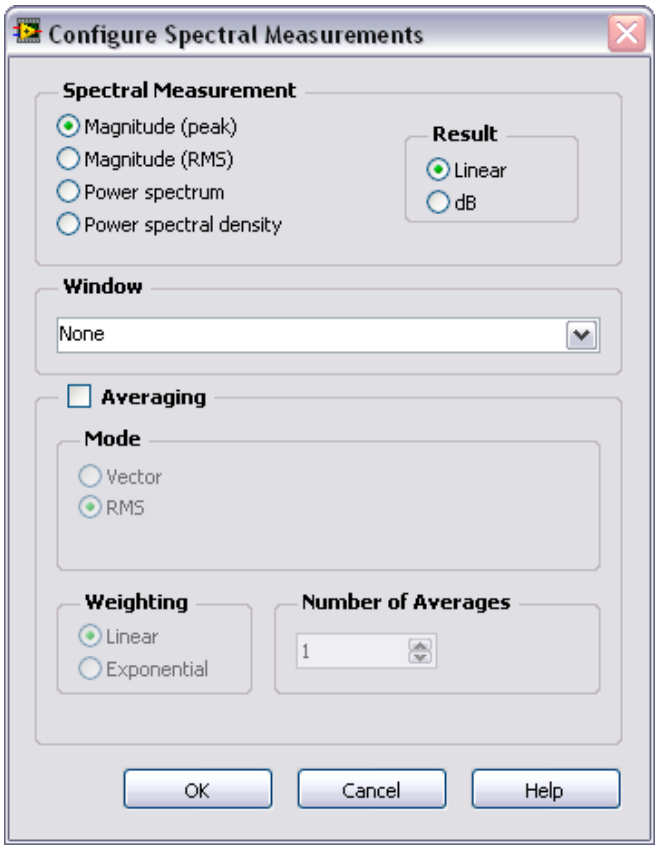
Cu dublu click pe acesta se configureaza ca in figura urmatoare:



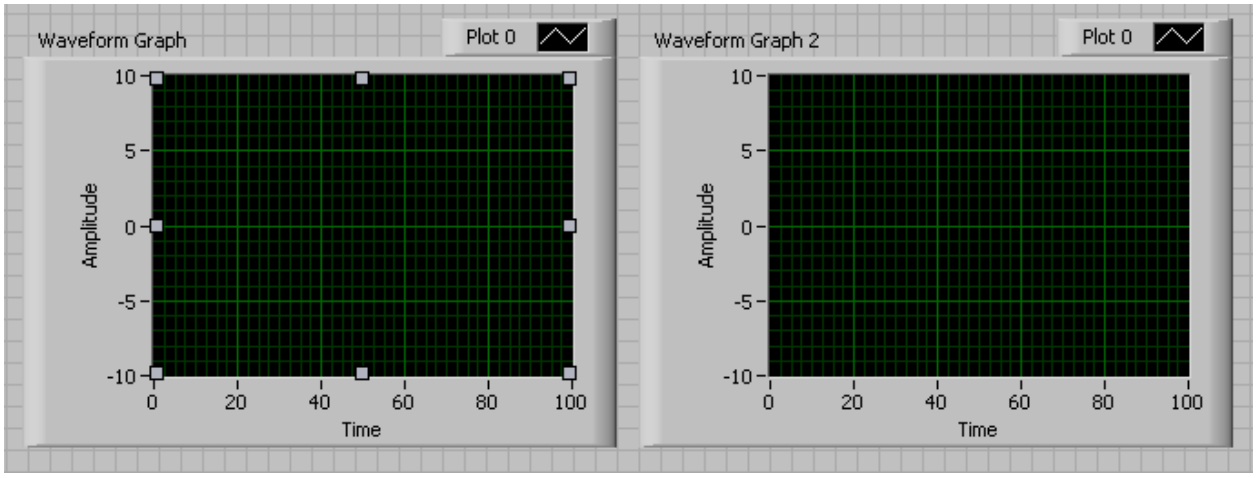
Pentru a observa caracteristicile de frecventa ale semnalului generat plasati un VI Spectral measurements (din Functions – Signal Processing – Frequency domain – Spectral measurements):



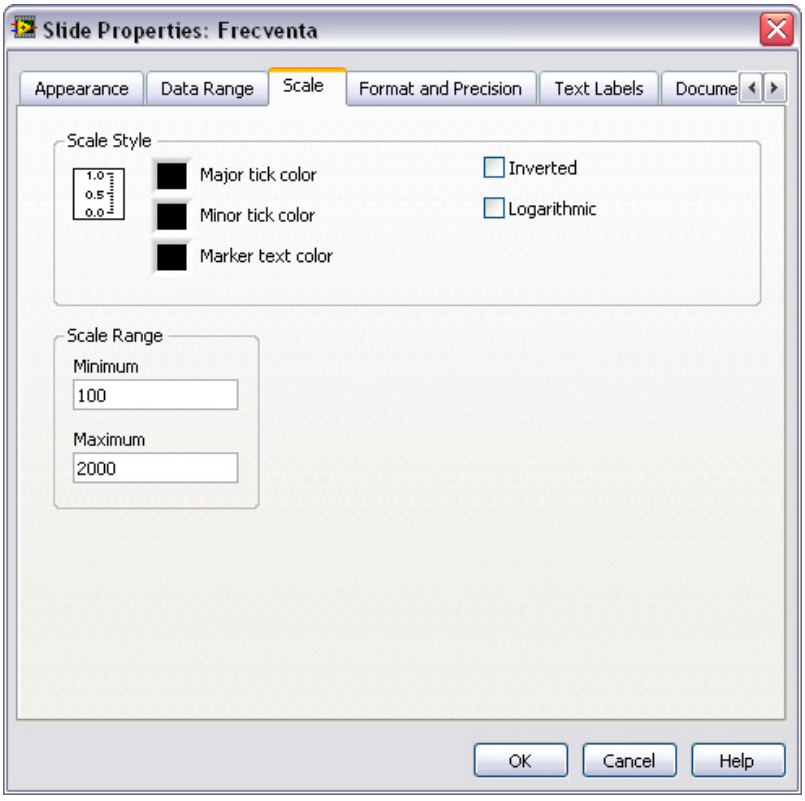
Configurati-l cu dublu click pe acesta in modul urmator:



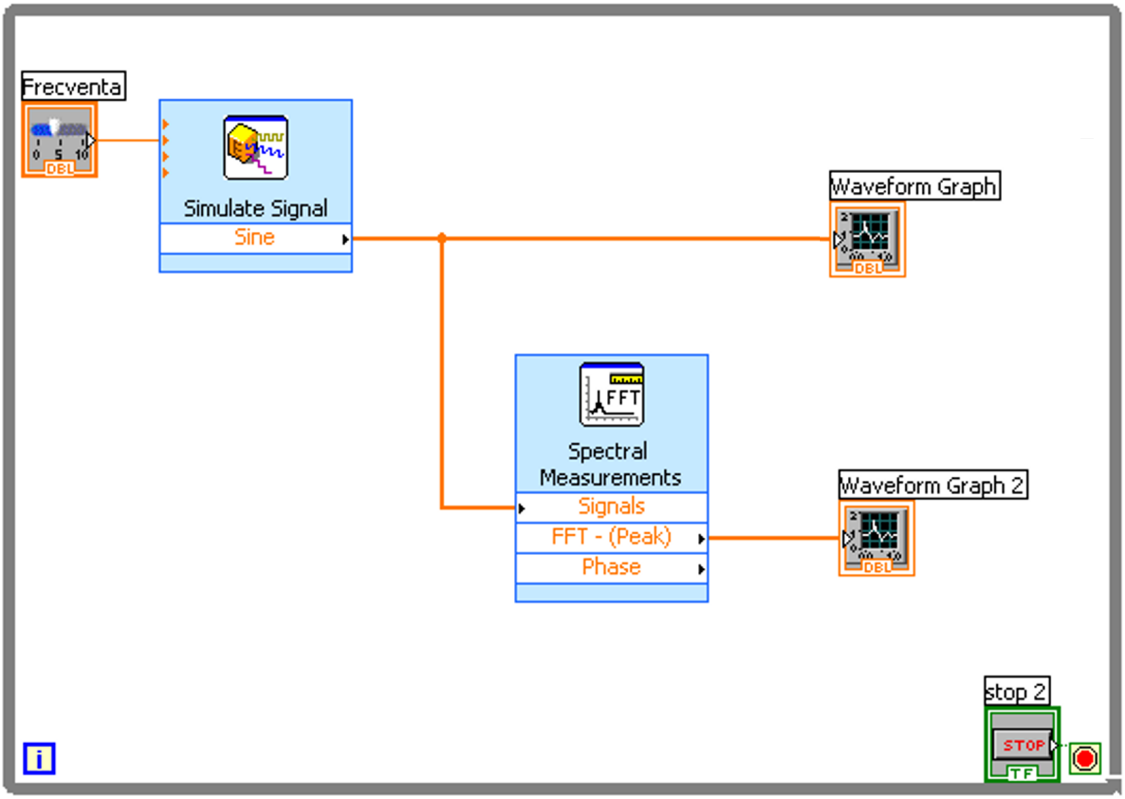
Treceti in panoul frontal si adaugati din Controls – Graph doua VI-uri Waveform Graph.



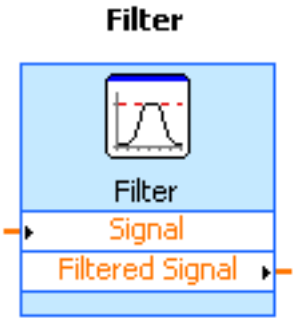
Adaugati un control de tip Horizontal Pointer Slide ce va fi folosit pentru modificarea frecventei. Cu click dreapta pe acesta alegeti Properties si configurati-l in modul urmator:



Realizati conexiunile ca in figura urmatoare si inchideti totul intr-o bucla while pentru ca acest program sa ruleze incontinuu:

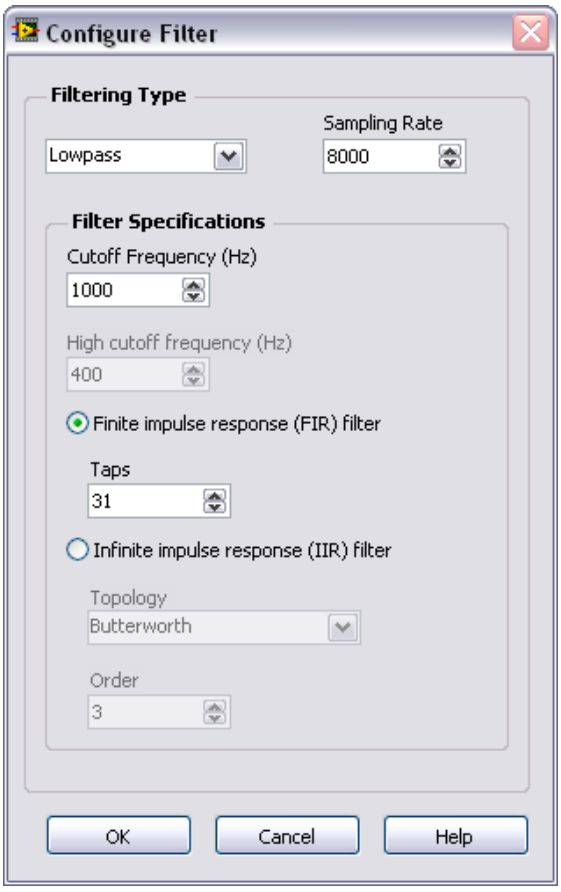


Se va modifica VI-ul precedent si vom adauga un bloc de filtrare digitala. Pentru aceasta vom folosi VI-ul Filter din Functions – Signal processing – Filters



Processes signals through filters and windows.

In acest exemplu vom implementa un filtru trece-jos, cu frecventa de taiere de 1000Hz. Parametrii filtrului pot fi modificati cu dublu click pe VI-ul acestuia:



Rulati acest VI. Comparati semnalul de iesire cu cel de intrare la diferite frecvente.

Modificati semnalul in unul dreptunghiular.